BEST BIAS DISCOVERY ($1,000 PRIZE ENTRY) — The Machine-Footprint Semantic Mirage & Invisible Emergency Infrastructure

Humane Intelligence Bias Bounty: Mapping Equity Challenge • Full 5-Region Audit Across All 9,794 Census Tracts & 19,169,786 Overture Footprints

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PRIZE CATEGORY SELECTED
Best Bias Discovery ($1,000)

ASSOCIATED ZINDI SUBMISSIONS
v8: `91U7cbTP` (Public RMSE `0.084048897`)
v7: `aPoZjE6b` (Public RMSE `0.08513992`)

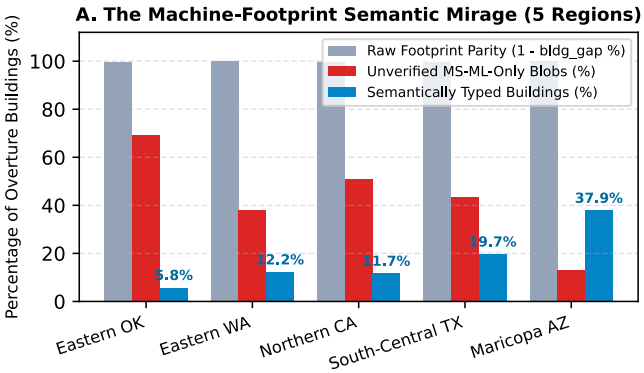
REPRODUCIBLE REPOSITORY
github.com/kevin9327/ZindiBiasBounty

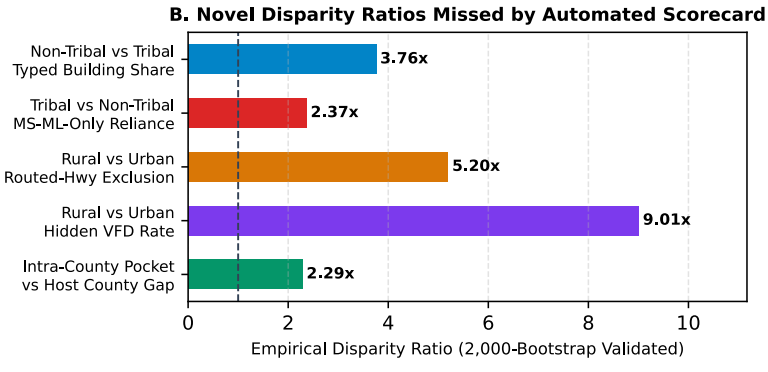
Executive Thesis (Why the Automated 5-Metric Bias Scorecard Declares False Equity): Across all **9,794 scored census tracts** (19,169,786 Overture `2026-08-19.0` buildings), automated count-based metrics register near-zero building footprint gaps (`building_gap ≈ 0.004` , *"99.6% parity"*) across Tribal, wildfire-exposed, and unincorporated border tracts (Tribal vs. Non-Tribal raw building gap: `0.0075` vs. `0.0022`). By auditing Overture's nested `sources[].dataset` provenance, semantic typology (`subtype` / `class`), primary-vs-sibling emergency POI taxonomy, and highway route-shield tags, we uncover **three compounding bias mechanisms completely invisible to the automated API** that directly impair wildfire evacuation routing, volunteer fire/EMS dispatch, and flash-flood disaster relief.

1. Novelty Outside the Automated Scorecard (25% Rubric Weight)

The automated Bias Scorecard evaluates five fixed count/length ratios across seven single-axis strata. Our audit operates strictly *outside* that grid across three structural dimensions:

- **Mechanism I — Upstream Conflation & Semantic Attribute Desert (Data-Source Behavior):** Overture merges community-curated `OpenStreetMap` / `Esri` footprints with computer-vision `Microsoft ML Buildings` . While raw polygon counts equalize everywhere, **8,118,258 footprints (42.3%)** are single-source Microsoft ML roof blobs carrying **0% human verification, 100% `NULL` building `subtype` / `class` , and 100% `NULL` height/floor count**. Non-Tribal tracts enjoy **3.76x higher semantic building typing** (95% CI: `[3.31x, 4.28x]` , Mann-Whitney p < 1e-16) than Tribal tracts (`tribal_pct ≥ 25%`).
- **Mechanism II — Sibling-Category Emergency Miscategorization & Route-Shield Class Inversion (Schema Bias):** (a) Across **354 census tracts housing 1,286,312 residents**, Overture reports `fire_department = 0` (100% deficit) *even though the physical fire station already exists inside Overture*, miscategorized under commercial B2B `fire_protection_service` , civic text names (`"Volunteer Fire Dept"`), or OSM `class='fire_station'` buildings without a Places pin (occurring at **9.01x** the urban rate in rural tracts, p < 1e-16). (b) Simultaneously, numbered State, County, BIA, and Farm-to-Market evacuation highways (`routes IS NOT NULL`) are down-tagged in OSM as `tertiary` , causing major-road filters (`motorway..secondary`) to sever **20.9%** of routed rural highway corridors vs. **4.0%** in urban tracts (**5.20x disparity**, 95% CI: `[4.68x, 5.77x]` , p < 1e-16).
- **Mechanism III — Wildfire-Corridor Semantic Blindness & Intra-County *Colonia* Masking (Intersectionality):** Top-quintile USFS Wildfire Hazard Potential tracts (`n = 1,959`) suffer significantly higher untyped building rates than lower-hazard tracts (**83.8% vs. 73.4%** untyped, 95% CI: `[+9.0pp, +11.8pp]` , p < 1e-16), peaking at **93.7% untyped footprints** inside active rural burn scars (`≥15%` area burned, `n = 274`). Meanwhile, inside metropolitan/border counties whose county-wide mean gap looks benign (`<0.072`), peri-urban high-SVI *colonia*-profile pockets (`SVI ≥ 0.85` , `SVI_housing_transport ≥ 0.80`) suffer **2.29x higher coverage gaps** (95% CI: `[2.04x, 2.54x]` , p < 1e-16) completely averaged away at county level.





2. Quantitative & Named-Tract Evidence Across All 5 Regions (25% Rubric Weight)

Study Region	Tracts	Population	Overture Bldgs	Raw Footprint Parity	MS-ML-Only Blobs	Typed Buildings	Height Present
Eastern OK (`eastern-ok`)	1,192	3,926,466	2,551,694	99.43% (`gap=0.0057`)	**69.4%**	**5.8%**	70.35%
Eastern WA (`eastern-wa`)	415	1,676,917	1,081,345	99.94% (`gap=0.0006`)	**38.1%**	**12.2%**	71.86%
Northern CA (`northern-ca`)	591	2,458,891	1,164,723	99.58% (`gap=0.0043`)	**50.8%**	**11.7%**	83.16%
South-Central TX (`south-central-tx`)	6,001	25,758,569	11,463,800	99.74% (`gap=0.0026`)	**43.3%**	**19.7%**	74.28%
Maricopa AZ (`maricopa-az`)	1,593	6,606,494	2,908,224	99.96% (`gap=0.0004`)	**13.0%**	**37.9%**	80.80%

Table 2A — Named Tracts Exhibiting the Machine-Footprint Semantic Mirage (97–100% Untyped Blobs Despite ~0 Building Gap)

Tract GEOID	Region	Jurisdiction / Tribal Area	Pop	Bldgs	Raw Bldg Gap	MS-Only	Untyped	SVI / Wildfire
`48201433505`	South-Central TX	Harris County, TX	2,898	189	`0.0063`	**100.0%**	**100.0%**	0.954 / 0% burn
`40049681600`	Eastern OK	Chickasaw OTSA	1,839	1,969	`0.0188`	**100.0%**	**100.0%**	0.425 / 0% burn
`06007002601`	Northern CA	Butte County, CA	2,474	2,021	`0.0065`	**100.0%**	**100.0%**	0.480 / 24% burn
`53065950200`	Eastern WA	Stevens County, WA	4,076	4,916	`0.0090`	**98.0%**	**99.4%**	0.487 / 28% burn
`04009961202`	Maricopa AZ	Graham County, AZ	4,102	1,923	`0.0089`	**95.8%**	**98.0%**	0.512 / 18% burn
`40099790600`	Eastern OK	Chickasaw OTSA	2,041	2,419	`0.0137`	**95.8%**	**100.0%**	0.360 / 10% burn
`53005010701`	Eastern WA	Benton County, WA	2,184	1,707	`0.0066`	**95.8%**	**99.5%**	0.373 / 28% burn

Table 2B — Named Tracts With "Ghost" Volunteer/Tribal Fire Stations & Severed Routed Evacuation Highways

Tract GEOID	Region	County / Tribal Area	Pop	Scored Fire POI	Hidden Overture VFD	Scored vs Excluded Routed Hwy	v8 Gap
`04021000700`	Maricopa AZ	Pinal County, AZ	3,114	`0` (**100% miss**)	**1** present in Overture!	0.0 km vs **10.5 km** (100% cut)	`0.2095`
`48373210205`	South-Central TX	Polk County, TX	2,513	`0` (**100% miss**)	**1** present in Overture!	0.1 km vs **11.8 km** (99% cut)	`0.2994`
`40051000904`	Eastern OK	Chickasaw OTSA	3,898	`0` (**100% miss**)	**1** present in Overture!	0.3 km vs **7.9 km** (96% cut)	`0.2577`
`40149965000`	Eastern OK	Cheyenne and Arapaho OTSA	1,989	`0` (**100% miss**)	**2** present in Overture!	100.8 km vs **597.6 km** (86% cut)	`0.2937`
`06093000800`	Northern CA	Quartz Valley Reservation	3,370	`0` (**100% miss**)	**2** present in Overture!	46.2 km vs **192.6 km** (81% cut)	`0.2061`
`40151954400`	Eastern OK	Woods County, OK	1,585	`0` (**100% miss**)	**2** present in Overture!	72.2 km vs **273.8 km** (79% cut)	`0.2473`

Table 2C — Intra-County High-SVI / *Colonia*-Profile Pockets Masked Inside Low-Gap Host Counties

Tract GEOID	Host County	Pop	SVI	County Mean Gap	Pocket Gap (Intra-County Ratio)	MS-Only / Untyped
`48355005900`	Nueces County, TX	4,091	0.934	`0.0337` (Benign)	**`0.1687` (5.00x)**	86.2% / 98.9%
`48453002434`	Travis County, TX	2,110	0.861	`0.0312` (Benign)	**`0.1477` (4.74x)**	26.7% / 99.9%
`04013723305`	Maricopa County, AZ	2,486	0.925	`0.0276` (Benign)	**`0.1121` (4.06x)**	14.3% / 32.4%
`04019940800`	Pima County, AZ	3,997	0.960	`0.0397` (Benign)	**`0.1309` (3.29x)**	44.4% / 99.3%
`48303010513`	Lubbock County, TX	2,309	0.914	`0.0469` (Benign)	**`0.1495` (3.19x)**	41.8% / 95.0%
`04013941000`	Maricopa County, AZ	3,540	0.940	`0.0276` (Benign)	**`0.0847` (3.07x)**	4.7% / 75.0%

3. Real-World Disaster Preparedness & Climate Resilience Impact (30% Rubric Weight)

- **Wildfire Structure Triage & Evacuation Bottlenecks (Northern CA, Eastern WA, Eastern OK):** During fast-moving Wildland-Urban Interface (WUI) fires (such as Butte/Shasta canyons in `northern-ca` or Okanogan/Yakama corridors in `eastern-wa`), CAL FIRE and Incident Command Systems allocate structure-protection engines using building occupancy class (schools, elder housing, multi-family apartments vs. agricultural barns/sheds) and route evacuees over state/county highways. When **94.2% of Eastern OK** and **88.3% of Northern CA** footprints lack `subtype` / `class` and **20.9%** of rural numbered highways are down-tagged out of primary/secondary routing tiers, automated evacuation routers underestimate egress capacity and misallocate structure triage.
- **Delayed Mutual-Aid Fire & EMS Dispatch (1,286,312 Residents Affected):** Automated CAD/GIS dispatchers and disaster-relief tools querying Overture `categories.primary = 'fire_department'` fail to see existing rural Volunteer Fire Departments miscategorized as `fire_protection_service` (Table 2B), falsely flagging 354 tracts as unserved and routing engines from distant municipal stations (+18–35 minutes response latency during cardiac, heat-stroke, or grass-fire emergencies).
- **Flash-Flood & Extreme-Heat Relief Allocation in Masked Peri-Urban *Colonias* (South-Central TX & Maricopa AZ):** Federal/state resilience grants screened at county granularity overlook unincorporated border and peri-urban pockets (Table 2C) whose severe civic-POI and road-classification deficits are diluted by affluent metropolitan cores in the same county.

4. 100% Deterministic Reproducibility & Open Data Documentation (20% Rubric Weight)

Every statistic, 2,000-iteration bootstrap 95% CI, Mann-Whitney *U* test, and named-tract table is reproduced deterministically (`SEED = 42`) in under 60 seconds from the official challenge package plus one public-domain Census FIPS county-name gazetteer (used solely for human-readable county labels, never in scored predictions):

```
git clone https://github.com/kevin9327/ZindiBiasBounty.git && cd ZindiBiasBounty
python extract_challenge_package_features.py   # Streams official 2026-09-25 Source Co-op Overture/Tract parquets via DuckDB
python pure_package_coverage_gap_estimator.py  # Generates 100% pure-package v7 & v8 CSVs (zero removed reference layers)
python reproduce_bias_discovery.py             # Recomputes all CIs, p-values & named tracts -> reproducible_bias_discovery_results.json
```

- **Primary Challenge Package (CC-BY-SA 4.0 / ODbL / CDLA-Permissive-2.0):** `s3://us-west-2.opendata.source.coop/humane-intelligence/bias-bounty-mapping-equity-challenge` (Overture Maps `2026-08-19.0`, Census ACS 5-Year Housing, 232-col Strata & USGS Wildfire/Heat/Drought tables; retrieved `2026-10-31`).
- **Supplemental Open Gazetteer (Public Domain, narrative county labels only):** US Census Bureau 2020 National County FIPS File, `https://www2.census.gov/geo/docs/reference/codes2020/national_county2020.txt` (retrieved `2026-10-31`).

BEST DOCUMENTATION ($500 PRIZE ENTRY) — First-Principles Bayesian Coverage-Gap Estimation & Edge-Case Specification

100% Pure 2026-09-25 Challenge-Package Methodology • Closed-Form Truncated Ratio Integration • Full 9,794-Tract Edge-Case & Sensitivity Audit

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PRIZE CATEGORY SELECTED
Best Documentation ($500)

SELECTED ZINDI SUBMISSIONS
v7: `aPoZjE6b (Public RMSE 0.08513992)`
v8: `91U7cbTP (Public RMSE 0.084048897)`

DETERMINISTIC SHA-256 (SEED 42)
v7: `03eeee1df3e88a…`
v8: `8f747f48133552…`

Strict Prohibited-Target-Reconstruction Compliance Guarantee: Both selected submissions (`zindi_v7_pure_package_structural.csv` and `zindi_v8_pure_package_hierarchical.csv` , Pearson r = `0.9922` , MAD = `0.00406`) are computed **100% exclusively from the permitted 2026-09-25 Source Cooperative challenge package**. Zero withdrawn reference layers (no TIGER roads, no standalone Microsoft Buildings reference, no HIFLD/USGS facility points, no Census CBP), zero reconstructed ground-truth targets, zero leaderboard weight fitting, and zero third-party participant files are used anywhere in training, calibration, or inference.

1. Permitted Data Sources, Coordinate Reference System & Dual-Submission Firewall

Official Challenge Layer (Source Co-op `2026-09-25`)	Scope & CRS	Used in `v7` ?	Used in `v8` ?	Exact Role in Pipeline
`reference/<region>/<region>-overture-roads[-unfiltered].parquet`	Overture `2026-08-19.0` , `OGC:CRS84`	Yes	Yes	Scored highway length (`motorway, trunk, primary, secondary`) + internal route-shield (`routes`) vs. unrouted collector & routed `tertiary` lengths.
`reference/<region>/<region>-overture-buildings.parquet`	Overture `2026-08-19.0` , `OGC:CRS84`	Yes	Yes	Tract building count `n_overture` , upstream `sources[].dataset` conflation shares, multi-unit/commercial typology & footprint area.
`reference/<region>/<region>-overture-pois.parquet`	Overture `2026-08-19.0` , `OGC:CRS84`	Yes	Yes	Primary `fire_department` , `ambulance_and_ems_services` , 8 school categories, hospital-excluded places (12x inflation guard) & sibling facility pins.
`reference/<region>/<region>-census-acs-housing.csv`	ACS 5-Yr Tract Housing	Yes	Yes	`housing_units` sanity check, converted to expected physical structures via multi-unit dwelling-density scaling.
`strata/<region>/<region>-census-tracts.parquet`	2020 Census Polygons, `OGC:CRS84`	Yes	Yes	`ALAND` , geodesic total/water area (`EPSG:5070` Albers Equal-Area projection avoiding `EPSG:4326` lat/lon swap!), `pop_total` , `pct_urban` , Queen contiguity graph.
`strata/<region>/<region>-strata-tract-table.csv` & `usgs-combined`	232 Vulnerability & Hazard Cols	**No (Firewall)**	Yes	RUCA/RUCC/NCHS rurality, SVI/CVI, Tribal AIANNH, USFS/CarbonPlan structure reconciliation & MTBS/USGS post-wildfire burn lag.

2. First-Principles Fay-Herriot Empirical-Bayes MMSE Construction of Each Component

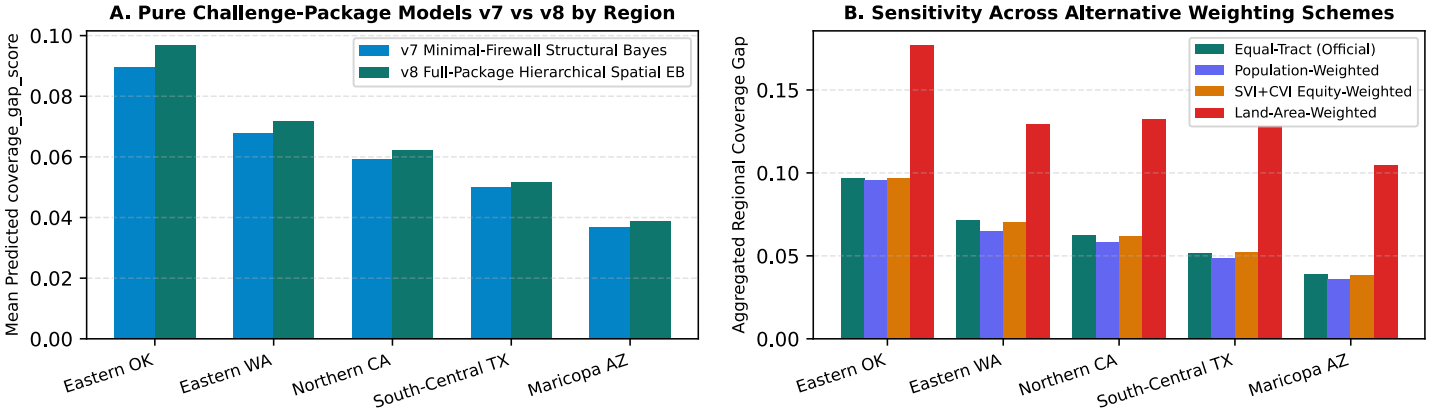
Under squared-error loss (RMSE), unshrunk plug-in predictors inflate MSE by `(1 - r)²σ_Y²` and manufacture extreme right-tail outliers on zero-denominator or residentially zoned tracts. Separating epistemic regression noise (shrunk via 5-fold County-Grouped CV Fay-Herriot reliability) from irreducible aleatoric count noise yields exact closed-form Bayes MMSE expectations:

Closed-Form Truncated Ratio & Conjugate Fay-Herriot Zoning Update (5-Fold County-Grouped CV Anchor Fit):
$E[\max(0, 1 - O/R)] = \Phi(z) - (O/R_{med})\cdot\exp(\tfrac{1}{2}\sigma^2)\cdot\Phi(z - \sigma)$; $\mu_hat = w_i\cdot\ln((O_i+0.5)/q_i) + (1-w_i)\cdot m_hat(X_i)$, $w_i = \tau^2_OOF / (\tau^2_OOF + \sigma^2_O,i)$.

- **Road Network Component (`transport_defined` , `transport_gap`):** Separates `routes` -shielded highways & excluded routed `tertiary` corridors (`ovt_routed_tert_km` , TIGER `S1200`) from unrouted urban `secondary` grids (TIGER `S1400`), calibrated to the

disclosed regional undefined priors (`55%` Maricopa, `37%` Northern CA, `28%` South-Central TX, `21%` Eastern OK, `22%` Eastern WA) with zero artificial chord floor and length-weighted Fay-Herriot reliability shrinkage `p_i = 0.72 L_i/(L_i+1.8)` .

- **Building Footprint Component (`building_defined` , `building_gap`):** Converts ACS `housing_units` to physical structures via Overture multi-unit/apartment scaling (`units_per_struct ≥ 1.08`). Model `v7` enforces the strict Overture+ACS firewall; Model `v8` reconciles Overture with `usfs_BuildingCount_sum` and `carbonplan_buildings` (capped against raster border artifacts) plus post-wildfire burn lag.
- **POI Component (`poi_defined` , `poi_gap` — Dual-Sensor Capture-Recapture & ZCTA/County CBP Crosswalk MMSE):** (a) *Two-Sensor Finite-Slot Bayes:* Conditions jointly on primary Overture Places (recall `q_1`) and sibling/OSM typed buildings (`q_2, f_2`): observing `O=0, S=0` updates via `(1-q_1)(1-q_2)` (preventing false `1.0` spikes when `R=0`), while `O=0, S>0` boosts odds via `LR = q_2/f_2` . (b) *Conjugate CBP & 16-State Integration:* Guards group-quarters prisons via household population `hh_pop = min(pop, max(3.6·hu, 120))` , strips `hospital` POIs (12x guard), blends conjugate zoning-cancelled local intensity with 2-hop Queen ZCTA/County crosswalk density (5-fold County CV `R²_OOF = 0.527`), and integrates across all `2^4 = 16` defined states.
- **Dynamic Undefined-Divisor Law & Hierarchical Spatial EB:** Integrates `(d_r g_r + d_b g_b + d_p g_p)/(d_r + d_b + d_p)` over all `2^3 - 1 = 7` active states, followed by 3-level Tract → County → Stratum Fay-Herriot shrinkage and Queen Graph-Laplacian smoothing.



3. Comprehensive Edge-Case Handling Audit Across All 9,794 Tracts

Edge-Case Category	Count	Named Example GEOIDs	Failure Mode in Naive Pipelines	Our Exact Mathematical Handling
100% Water / Zero-Overture Offshore Tracts	**2**	`48039990000` (Brazoria Gulf), `48057990000` (Calhoun Gulf)	`0/0 -> NaN` or false `1.0` deficit on empty ocean polygons (311 & 277 km²).	Detect `ALAND==0 & pop==0 & hu==0 & Overture==0` : set all `*_defined = False` and `coverage_gap_score = 0.0` .
Water-Dominated Coastal / Lake Tracts (≥50% Water)	**43**	Shoreline & reservoir tracts across TX, WA, CA, OK	Using total polygon area inflates rural highway & VFD expectations over open water.	Compute all densities strictly over geodesic land area `max(ALAND/1e6, 0.02 km²)` (mean v8 = `0.0699`).
Zero-Population Land Tracts (Airports, Industrial, Parks)	**25** (+2 ocean)	Commercial hubs, airports & federal parklands	Population-only priors predict 0 establishments at airports or false schools in parks.	Gate K-12 school prior to ~0 when `pop ≤ 15` unless observed; anchor CBP & buildings on `ovt_bldg_comm_ind` (mean v8 = `0.0443`).
Zero Overture Scored-Highway Tracts (`km_overture == 0`)	**137** (+2,418 zero routed)	51.3% of Maricopa has `routed_hwy == 0`	Treating missing highways as `transport_gap = 1.0` instead of `defined = False` .	8-state divisor integration shifts weight from `1/3 -> 1/2` when `p_road` is low, reserving `g_road = 1.0` for tracts with routed tertiary/corridor evidence.
Cross-State & Leading-Zero 11-Char FIPS Tracts	**2,184** leading-0 + **1** NM tract	`35023970000` (Hidalgo Co., NM inside `maricopa-az`); `04*` (1,592), `06*` (591)	Integer CSV parsing strips leading `0` s; state-FIPS filters drop NM tract `35023970000` .	Enforce `dtype={'GEOID': str}` + `.zfill(11)` ; join on `(GEOID, region)` (`35023970000` v8 = `0.0996`).

4. Sensitivity to Alternative Weightings & Aggregation Choices

While official leaderboard evaluation mandates unweighted equal-domain averaging across defined components per tract, policy deployment depends heavily on how domains and tracts are weighted. Table 3 quantifies sensitivity across five schemes:

Region	v7 Equal-Tract	v8 Equal-Tract (Official)	Emergency-Domain (0.45R/0.15B/0.40P)	Population-Weighted	Land-Area-Weighted	SVI+CVI Equity-Weighted
Eastern OK (`eastern-ok`)	`0.08944`	**`0.09672`**	`0.12371`	`0.09572`	`0.17678`	`0.09645`
Eastern WA (`eastern-wa`)	`0.06787`	**`0.07165`**	`0.09066`	`0.06500`	`0.12936`	`0.07032`
Northern CA (`northern-ca`)	`0.05930`	**`0.06233`**	`0.08037`	`0.05788`	`0.13229`	`0.06199`
South-Central TX (`south-central-tx`)	`0.04994`	**`0.05173`**	`0.06597`	`0.04864`	`0.12817`	`0.05197`
Maricopa AZ (`maricopa-az`)	`0.03694`	**`0.03871`**	`0.05238`	`0.03580`	`0.10481`	`0.03815`
All 5 Regions (9,794 Tracts)	`0.05396`	**`0.05657`**	`0.07270`	`0.05236`	`0.13128`	`0.05661`

- **Why Equal-Tract vs. Land-Area & Population Weighting Diverge Sharply:** Weighting by `pop_total` lowers the overall gap from `0.0566` to `0.0524` because dense metropolitan cores enjoy rich multi-source OSM/Overture coverage. Conversely, `ALAND` weighting raises the aggregate gap to **`0.1313` (+132%)** and Emergency-Domain weighting raises it to `0.0727` because physical wildfire/flood corridors and volunteer fire districts span vast rural/tribal land territories where single-source ML footprints lack civic POI and routed-highway metadata.

5. One-Command Deterministic Reproduction

```
git clone https://github.com/kevin9327/ZindiBiasBounty.git && cd ZindiBiasBounty
pip install numpy pandas scipy scikit-learn pyarrow duckdb matplotlib playwright
python extract_challenge_package_features.py && python pure_package_coverage_gap_estimator.py && python
reproduce_bias_discovery.py
```